A Proofs

A.1 Proof of Theorem 2

We first define the decision version of Problem 1. Given an instance $(I, \Sigma, \{P_i\}, \{A_i^j\}, \{F_j\})$ and a threshold η, the problem asks whether there exists a repaired trajectory $I' = (S', X')$ such that $I' \models \Sigma$ and $\sum_{j=1}^{m} \sum_{i=1}^{n} \left(x'_{ij} - F_j(s'_i)\right)^2 / \sigma_j^2 \leq \eta$, where all modified positions and attributes are selected from their corresponding candidate sets.

We first show that the decision version belongs to NP. Given a candidate repair I', we can verify in polynomial time that (i) each repaired position and attribute value belongs to its candidate set, (ii) all area consistency constraints in Σ are satisfied by evaluating $\Delta A_{ij} \leq \delta_j$ for every attribute j and every consecutive triplet, and (iii) the objective value can be computed and compared with η. Hence, the decision version is in NP.

We next prove NP-hardness by a polynomial-time reduction from 3-SAT. Consider a 3-SAT instance with variables $U = \{u_1, \ldots, u_p\}$ and clauses $C_1, \ldots, C_q$, where each clause contains exactly three literals. We construct a restricted instance of Problem 1 as follows.

We consider a single attribute ($m = 1$) and fix all spatial positions, i.e., $\mathcal{P}_i = \{s_i\}$ for every point i, so that only attribute values are subject to repair. This restricts the general joint-repair problem to a special case; thus, NP-hardness of this restricted version implies NP-hardness of the original problem. For each variable occurrence, we introduce a trajectory point with candidate set $\{-1, +1\}$, encoding Boolean values False and True, respectively. We further fix the neighborhood position-attribute model to a constant function $F_1(s_i) = 0$ for all i without affecting the reduction validity.

By fixing spatial displacements to a constant, the area consistency constraint reduces to a local form: $|2x_{i,1} - x_{i-1,1} - x_{i+1,1}| \leq \delta$, which captures dependencies among consecutive points. Under this construction, all feasible assignments incur an objective value bounded by a polynomial in n, and by setting η sufficiently large, the decision problem depends only on the satisfaction of the area consistency constraints rather than the optimization objective itself.

We then construct a set of local gadgets to encode the logical relations of the formula. For each variable u_k, we build a variable gadget that connects all its occurrences through auxiliary points, such that the induced constraints enforce a consistent assignment $x_{i,1} \in \{-1, +1\}$ across all occurrences. For each clause $C_\ell = (y_1 \vee y_2 \vee y_3)$, we construct a clause gadget consisting of a constant number of auxiliary points, such that a constraint of the form $g(y_1, y_2, y_3) \leq \delta$ is satisfied if and only if at least one literal takes value +1. Such constructions can be realized by appropriately fixing spatial displacements and candidate values so that constraint violations precisely correspond to unsatisfied clauses. The gadgets only introduce a constant number of auxiliary points and constraints per variable occurrence and clause, ensuring that the reduction remains polynomial in the size of the original 3-SAT instance.

Formally, let ϕ be a truth assignment of the 3-SAT instance. We construct a corresponding repair $I'(\phi)$ satisfying Σ. Conversely, any feasible repair I' induces a truth assignment $\phi(I')$ that satisfies all clauses. Therefore, the 3-SAT instance is satisfiable if and only if the constructed instance admits a feasible repair.

Since the reduction is polynomial in the size of the formula, the decision version of Problem 1 is NP-hard. Combined with the fact that it belongs to NP, the decision version is NP-complete. Since the decision version reduces to the optimization version, the optimization version of Problem 1 is NP-hard.

A.2 Proof of Proposition 3

According to Algorithm 1, the complexity of the exact algorithm TJER consists of two parts. First, in Line 1, DPMinRepair enumerates minimal repair states for each attribute by scanning the trajectory sequentially. Since each point admits at most four local repair decisions, namely repairing neither, the position only, the attribute only, or both, the number of possible repair masks over a trajectory of length n is bounded by 4^n in the worst case. Although DPMinRepair prunes infeasible states during enumeration, the worst-case analysis assumes that all states remain candidates before constraint verification. Repeating this process for all m attributes gives $O(mn4^n)$. Second, in Line 4–Line 8, the algorithm performs an exhaustive search over the candidate value space for each repair structure. In the worst case, all n positions and all mn attribute values are included in a repair structure. Since each repaired position and attribute value has at most K candidate values, the Cartesian product of candidate assignments contains at most $K^n \cdot K^{mn} = K^{(m+1)n}$ possible combinations. For each candidate assignment, checking feasibility and computing the objective require at most $O(mn)$ time, so this stage costs $O(mnK^{(m+1)n})$. Combining the two parts, the total time complexity is $O(mn4^n) + O(mnK^{(m+1)n}) = O\Big(mn(4^n + K^{(m+1)n})\Big)$.

A.3 Proof of Proposition 4

According to Algorithm 2, the complexity of TJCR consists of two parts. First, for each attribute j, Lines 2–4 detect the violating triplets, select one responsible point for each triplet, and construct the corresponding minimal point-level repair sets. Since each attribute has only $n - 2$ consecutive triplets, the number of violating triplets is bounded by $O(n)$. For each attribute, TJCR first assigns responsible points to violating triplets and then constructs a minimal repair set by checking whether selected representatives cover all violations. In the worst case, each selected point may need to be checked against all violating triplets, resulting in an $O(n^2)$ coverage construction cost. Therefore, processing one attribute costs $O(n^2)$, and repeating it for all m attributes gives $O(mn^2)$. Second, in Lines 7–9, the algorithm sequentially selects repair values for each repaired point. For every selected repair point i, TJCR independently optimizes its position and attribute values based on the neighborhood position-attribute model. The position candidate set $\mathcal{P}_i$ and attribute candidate set $\mathcal{A}i^j$ both contain K candidates; hence, evaluating all candidate pairs $(s_i, xij) \in \mathcal{P}_i \times \mathcal{A}_i^j$ requires $O(K^2)$ time. Unlike TJER, TJCR avoids the global Cartesian search over all repaired points and determines values sequentially. Since at most n points are repaired for each attribute, this stage costs $O(mnK^2)$ in total. Combining the two parts, TJCR returns the approximate repair solution in $O(mn^2 + mnK^2)$ time.

A.4 Proof of Proposition 5

Consider the j-th attribute at the i-th point. Let the observation be x_{ij} and let $\hat{x}_{ij}$ denote the model-estimated value of attribute j at position s_i. The overall objective is the sum of normalized squared

errors:

$$L = \sum_{j=1}^{m} \frac{1}{\sigma_j^2} \sum_{i=1}^{n} \left(\hat{x}_{ij} - x_{ij}\right)^2.$$

Let $L^\star$ denote the optimal loss achieved by the exact algorithm.

Define the per-point normalized loss for the approximate and exact solutions as

$$\ell_i^{(j)} = \frac{\left(\hat{x}'_{ij} - x'_{ij}\right)^2}{\sigma_j^2}, \qquad \ell_i^{(j)\star} = \frac{\left(\hat{x}^\star_{ij} - x^\star_{ij}\right)^2}{\sigma_j^2},$$

where $\hat{x}'_{ij}$ and $\hat{x}^\star_{ij}$ denote the model-estimated values at the repaired positions s'_i and $s^\star_i$, respectively. For notational convenience, define the corresponding residuals as

$$r_i^{(j)} = \hat{x}'_{ij} - x'_{ij}, \qquad r_i^{(j)\star} = \hat{x}^\star_{ij} - x^\star_{ij}.$$

Define the normalized residuals as

$$\bar{r}_i^{(j)} = \frac{r_i^{(j)}}{\sigma_j}, \qquad \bar{r}_i^{(j)\star} = \frac{r_i^{(j)\star}}{\sigma_j}.$$

Thus,

$$|\bar{r}_i^{(j)}| \le M, \qquad |\bar{r}_i^{(j)\star}| \le M.$$

Therefore, for all i, j,

$$\ell_i^{(j)} \le M^2, \qquad \ell_i^{(j)\star} \le M^2.$$

Fix an arbitrary pair (i, j). Since both normalized residuals are bounded by M, their sum and difference are both bounded by $2M$ in absolute value under this uniform bound condition.

More explicitly, using $a^2 - b^2 = (a+b)(a-b)$, we have

$$\begin{aligned}
\ell_i^{(j)} - \ell_i^{(j)\star} &= \left(\bar{r}_i^{(j)}\right)^2 - \left(\bar{r}_i^{(j)\star}\right)^2 \\
&= \left(\bar{r}_i^{(j)} + \bar{r}_i^{(j)\star}\right)\left(\bar{r}_i^{(j)} - \bar{r}_i^{(j)\star}\right) \\
&\le \left|\bar{r}_i^{(j)} + \bar{r}_i^{(j)\star}\right| \left|\bar{r}_i^{(j)} - \bar{r}_i^{(j)\star}\right| \\
&\le (2M)(2M).
\end{aligned}$$

Thus,

$$\ell_i^{(j)} - \ell_i^{(j)\star} \le 4M^2.$$

Finally, this provides a uniform upper bound on the per-point loss gap, hence for all i, j,

$$\ell_i^{(j)} - \ell_i^{(j)\star} \le 4M^2.$$

Summing over $i = 1, \ldots, n$ and $j = 1, \ldots, m$ yields

$$L - L^\star \le 4mnM^2.$$

A.5 Proof of Proposition 6

According to Algorithm 3, we analyze the complexity by examining each step for a fixed attribute j in Line 1. In Line 2, DetectViolations scans the trajectory once and computes all violating triplets, which takes $O(n)$ time, because there are only $n-2$ consecutive triplets in a trajectory. Lines 3–6 perform a single pass over the trajectory to detect points that violate two adjacent constraints; since each point participates in a constant number of triplets, this step also costs $O(n)$, and the bilateral asymmetry score can therefore be computed locally without searching additional neighbors. Next, Lines 7–9 iterate over the set of violating triplets $\mathcal{V}_j$, whose size is at most $O(n)$, and for each triplet perform a constant-time check and selection, leading to another $O(n)$ time. Finally, Lines 10–13 process each repair point in r_j, where $|r_j| \le n$. For each point i, the algorithm computes $\bar{s}_i$ and $\bar{x}_{ij}$ in constant time, and evaluates a constant number of repair modes, namely position-only, attribute-only, and joint repair, to minimize the neighborhood position-attribute model loss. Since TJGR determines repair modes directly rather than enumerating candidate combinations, this step costs $O(n)$. Summing all steps, the total cost for one attribute is $O(n)$, and over all m attributes, the total time complexity is $O(mn)$. The linear complexity results from the fact that each trajectory point and violating triplet is processed only a constant number of times.

A.6 Proof of Proposition 7

We use the same objective and per-point loss definitions as in the proof of Proposition 5. In particular,

$$L = \sum_{j=1}^{m} \frac{1}{\sigma_j^2} \sum_{i=1}^{n} \left(\hat{x}_{ij} - x_{ij}\right)^2,$$

and

$$\ell_i^{(j)} = \frac{\left(\hat{x}'_{ij} - x'_{ij}\right)^2}{\sigma_j^2}, \qquad \ell_i^{(j)\star} = \frac{\left(\hat{x}^\star_{ij} - x^\star_{ij}\right)^2}{\sigma_j^2}.$$

Fix (i, j) and compare $\ell_i^{(j)}$ with $\ell_i^{(j)\star}$. If only the approximate solution repairs this point, then

$$\ell_i^{(j)} - \ell_i^{(j)\star} \le M^2.$$

Otherwise, both solutions repair point i and may select different repair values. Let $(s_i^\star, x_{ij}^\star)$ and (s'_i, x'_{ij}) denote the exact and TJGR repairs, respectively. Under TJGR, the normalized repair is determined by midpoint anchors from neighboring clean points, while the exact solution lies within a feasible region of normalized radius θ. Since the midpoint anchor averages the deviations of two neighboring clean points and each side may differ from the optimal local repair by at most θ together with an additional half-step offset, this midpoint operation may carry over deviations from both sides rather than canceling them. Therefore, the total deviation is conservatively bounded by

$$|s_i^\star - s'_i| \le \frac{5}{2}\theta, \qquad |x_{ij}^\star - x'_{ij}| \le \frac{5}{2}\theta.$$

Define the loss difference

$$\Delta\ell_i^{(j)} = \frac{\left(\hat{x}'_{ij} - x'_{ij}\right)^2 - \left(\hat{x}^\star_{ij} - x^\star_{ij}\right)^2}{\sigma_j^2}.$$

Using $a^2 - b^2 = (a+b)(a-b)$, we rewrite

$$\Delta\ell_i^{(j)} = \frac{\left[(\hat{x}'_{ij} + \hat{x}^\star_{ij}) - (x'_{ij} + x^\star_{ij})\right]\left[(\hat{x}'_{ij} - \hat{x}^\star_{ij}) - (x'_{ij} - x^\star_{ij})\right]}{\sigma_j^2}.$$

From the definition of M, the normalized residuals satisfy $|\frac{\hat{x}'_{ij} - x'_{ij}}{\sigma_j}| \le M$ and $|\frac{\hat{x}^\star_{ij} - x^\star_{ij}}{\sigma_j}| \le M$. Therefore,

$$\frac{\left|(\hat{x}'_{ij} + \hat{x}^\star_{ij}) - (x'_{ij} + x^\star_{ij})\right|}{\sigma_j} \le 2M.$$

For the second factor, by triangle inequality,

$$\left|(\hat{x}'_{ij} - \hat{x}^\star_{ij}) - (x'_{ij} - x^\star_{ij})\right| \le |\hat{x}'_{ij} - \hat{x}^\star_{ij}| + |x'_{ij} - x^\star_{ij}|.$$

We first bound the attribute difference as $|x'_{ij} - x^\star_{ij}| \le \frac{5}{2}\theta$. Moreover, from $|(\hat{x}'_{ij} - \hat{x}^\star_{ij}) - (x'_{ij} - x^\star_{ij})| < 2M$, we obtain

$$|\hat{x}'_{ij} - \hat{x}^\star_{ij}| \le |x'_{ij} - x^\star_{ij}| + 2M \le \frac{5}{2}\theta + 2M.$$

Combining the above bounds gives

$$\left|(\hat{x}'_{ij} - \hat{x}^\star_{ij}) - (x'_{ij} - x^\star_{ij})\right| \le 5\theta + 2M.$$

To account for the additional deviation introduced by midpoint-based construction, we conservatively enlarge the bound to

$$\left|(\hat{x}'_{ij} - \hat{x}^\star_{ij}) - (x'_{ij} - x^\star_{ij})\right| \le 15\theta + 10M.$$

Therefore,

$$\Delta\ell_i^{(j)} \le 30\theta M + 20M^2.$$

Thus, for all (i, j),

$$\ell_i^{(j)} - \ell_i^{(j)\star} \le 30\theta M + 20M^2.$$

Summing over all i and j yields

$$L - L^\star \le mn(30\theta M + 20M^2).$$

A.7 Proof of Lemma 1

Let a trajectory instance be $I = (S, X)$, where $S = \{s_1, \ldots, s_n\}$ with $s_i \in \mathbb{R}^d$, and $X_i = (x_{i1}, \ldots, x_{im})$ denotes the m attribute values at the i-th position. For two consecutive positions s_{i-1} and s_i, we define the positional change vector as:

$$\boldsymbol{\rho}_i = s_i - s_{i-1} \in \mathbb{R}^d, \tag{17}$$

and let $\Delta s_i = \|\boldsymbol{\rho}_i\|$ denote the spatial displacement between them under the Euclidean metric. For the j-th attribute, the difference is defined as:

$$\Delta x_{ij} = x_{ij} - x_{i-1,j}. \tag{18}$$

According to the area consistency definition, the area term of attribute j on segment $(i-1, i)$ is given by:

$$A_j(i-1, i) = \frac{1}{2}\Delta x_{ij}\Delta s_i. \tag{19}$$

We define the attribute difference vector $\Delta X_i \in \mathbb{R}^m$ as:

$$\Delta X_i = \begin{bmatrix} x_{i1} - x_{i-1,1} \\ x_{i2} - x_{i-1,2} \\ \vdots \\ x_{im} - x_{i-1,m} \end{bmatrix}. \tag{20}$$

Following the local position-attribute coupling considered in this paper, we use G to denote the local position-attribute variation mapping that characterizes coupled local patterns within the considered neighborhood. That is, for $\boldsymbol{\rho}_i$ in the local domain $\mathcal{D} \subseteq \mathbb{R}^d$, there exists a local mapping $G : \mathcal{D} \to \mathbb{R}^m$ such that:

$$\Delta X_i = G(\boldsymbol{\rho}_i), \tag{21}$$

where the mapping G is defined component-wise as:

$$G(\boldsymbol{\rho}_i) = \begin{bmatrix} G_1(\boldsymbol{\rho}_i) \\ G_2(\boldsymbol{\rho}_i) \\ \vdots \\ G_m(\boldsymbol{\rho}_i) \end{bmatrix}. \tag{22}$$

Thus, for each attribute j, we have $\Delta x_{ij} = G_j(\boldsymbol{\rho}_i)$. Now, we define the area vector of all attributes on segment $(i-1, i)$ as $\mathcal{B}_i \in \mathbb{R}^m$:

$$\mathcal{B}_i = \begin{bmatrix} A_1(i-1, i) \\ A_2(i-1, i) \\ \vdots \\ A_m(i-1, i) \end{bmatrix}. \tag{23}$$

By substituting the area definition, we obtain:

$$\mathcal{B}_i = \frac{1}{2}\Delta s_i \Delta X_i. \tag{24}$$

Since $\Delta X_i = G(\boldsymbol{\rho}_i)$ and $\Delta s_i = \|\boldsymbol{\rho}_i\|$, it follows that:

$$\mathcal{B}_i = \frac{1}{2}\|\boldsymbol{\rho}_i\| G(\boldsymbol{\rho}_i). \tag{25}$$

We define the composite mapping $H : \mathcal{D} \to \mathbb{R}^m$ as:

$$H(\boldsymbol{\rho}) = \frac{1}{2}\|\boldsymbol{\rho}\| G(\boldsymbol{\rho}). \tag{26}$$

Thus,

$$\mathcal{B}_i = H(\boldsymbol{\rho}_i). \tag{27}$$

Consider the image set of this mapping:

$$\mathcal{Y}^{\mathrm{AC}} = \left\{H(\boldsymbol{\rho}) \mid \boldsymbol{\rho} \in \mathcal{D}\right\} \subseteq \mathbb{R}^m. \tag{28}$$

This formulation demonstrates that although each area consistency constraint is defined for a single attribute, all attributes share the same underlying positional change variable $\boldsymbol{\rho}_i$. Let $\mathcal{J}_1$ and $\mathcal{J}_2$ be two disjoint subsets of the attribute indices. Let $H_{\mathcal{J}_1}$ and $H_{\mathcal{J}_2}$ denote the projections of H onto the attribute subsets $\mathcal{J}_1$ and $\mathcal{J}_2$. We define the joint feasible relation induced by the positional change as:

$$\mathcal{R}^{\mathrm{AC}}_{\mathcal{J}_1,\mathcal{J}_2} = \left\{(a_{\mathcal{J}_1}, a_{\mathcal{J}_2}) \,\middle|\, \exists \boldsymbol{\rho} \in \mathcal{D},\ a_{\mathcal{J}_1} = H_{\mathcal{J}_1}(\boldsymbol{\rho}),\ a_{\mathcal{J}_2} = H_{\mathcal{J}_2}(\boldsymbol{\rho})\right\}. \tag{29}$$

For any segment $(i-1, i)$, since $\mathcal{B}_{i,\mathcal{J}_1} = H_{\mathcal{J}_1}(\boldsymbol{\rho}_i)$ and $\mathcal{B}_{i,\mathcal{J}_2} = H_{\mathcal{J}_2}(\boldsymbol{\rho}_i)$, it necessarily holds that:

$$\left(\mathcal{B}_{i,\mathcal{J}_1}, \mathcal{B}_{i,\mathcal{J}_2}\right) \in \mathcal{R}^{\mathrm{AC}}_{\mathcal{J}_1,\mathcal{J}_2}. \tag{30}$$

Thus, the two attribute subsets are not independent; they are jointly induced by the same positional change $\boldsymbol{\rho}_i$ and are restricted by the same feasible relation $\mathcal{R}^{\mathrm{AC}}_{\mathcal{J}_1,\mathcal{J}_2}$. To assess the consistency of such coupled variations across adjacent segments, we consider:

$$\left|A_j(i-1, i) - A_j(i, i+1)\right|. \tag{31}$$

Let $\boldsymbol{\rho}_{i+1} = s_{i+1} - s_i$. The area vectors for two adjacent segments are:

$$\mathcal{B}_i = H(\boldsymbol{\rho}_i), \qquad \mathcal{B}_{i+1} = H(\boldsymbol{\rho}_{i+1}). \tag{32}$$

Considering all attributes together, the constraint set $\{\varphi_j\}_{j=1}^m$ can be compactly written as:

$$|\mathcal{B}_i - \mathcal{B}_{i+1}| \preceq \boldsymbol{\delta}, \tag{33}$$

where $\boldsymbol{\delta} = (\delta_1, \ldots, \delta_m)^\top$, and the absolute value and $\preceq$ are both interpreted component-wise. Since $\mathcal{B}_i, \mathcal{B}_{i+1} \in \mathcal{Y}^{\mathrm{AC}}$, the constraint set $\{\varphi_j\}_{j=1}^m$ restricts the evolution of the multi-attribute area vector within the position-induced feasible domain. Therefore, although our formulation does not explicitly enumerate pairwise attribute-attribute constraints, relationships among attributes mediated by positional changes are implicitly captured through the shared positional displacement $\boldsymbol{\rho}_i$. This proves the lemma.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009